# Algebra II with Trigonometry Chapter 5.3

**Olympiad Problems (GPT-5.4)**

Saved tutoring response with MathJax rendering. Original PDF file: 4401269_alg-2trig-h-chp-5-3-note-docx.pdf

## Problems

1. Find the exact value of

$$\sin 255^\circ + \cos 345^\circ + \tan 225^\circ.$$

2. An angle $\theta$ satisfies $\sin\theta + \cos\theta = \frac{1}{3}$, and $\theta$ lies in Quadrant II. Find the exact value of $\tan\theta$.

3. Without using a calculator, determine the sign and exact value of

$$\sec\left(-\frac{7\pi}{6}\right) + \csc\left(\frac{11\pi}{6}\right).$$

4. A point $P(x, y)$ lies on the terminal side of an angle $\theta$ in Quadrant III, and

$$x + y = -7, \qquad \frac{x}{y} = \cot\theta = \frac{3}{4}.$$

Find the exact values of $\sin\theta, \cos\theta, \tan\theta$.

5. Two sides of a triangle have lengths $13$ and $20$, and the included angle is $150^\circ$. Find the area of the triangle.

6. An observer is $5280$ ft from the launch pad of a rocket. At one moment the angle of elevation is $45^\circ$; one minute later it is $60^\circ$. Assuming the rocket rises straight up at constant speed during that minute, how many feet does it rise?